

LARAVEL MINI PROJECT: MIDTERM REQUIREMENTS

Course: Web Development / Web Systems

Document Type: Student Handout & Paper Documentation Guidelines

1. General Project Instructions

Choose one system topic from the approved list and build a simple working CRUD application in Laravel. Alongside your software demonstration, submit a short paper documentation (3 to 5 pages) explaining how your system works.

Key Reminders

- **Group Size:** as agreed upon last time
- **Deliverables:**
 - a. PDF Mini Project Paper Documentation (3–5 pages).
 - b. Live project demonstration and oral defense during the midterm examination schedule.
- **No Source Code Policy:** Do NOT paste raw code scripts into your paper. Instead, provide written descriptions, architectural diagrams, Data Dictionary tables, route lists, and annotated UI screenshots.

2. List of Approved Project Topics

Simple Information Systems

- **Book Information System:** Add, edit, view, and delete book records (ISBN, title, author, category, publication year, shelf status).
- **Barangay Information System:** Manage resident profiles, household classifications, purok/zone assignments, and basic clearance requests.
- **Employee Information System:** Record employee profiles, department designations, job positions, employment status, and base salary rates.

- **Product Inventory System:** Track product items, category taxonomies, stock quantities, supplier profiles, and reorder alerts.

Advanced Concepts

- **HRIS (Simple):** Basic employee directory, department rosters, shift schedules, and daily attendance logging.
- **Point of Sale (POS) System:** Product item catalog, cashier shopping cart, order checkout, stock level reduction, and receipt summaries.
- **Library System:** Book catalog, student borrower profiles, circulation desk logs (borrowing, return dates, overdue fines).
- **Simple Customer Relationship Management (CRM):** Customer contact records, inquiry tracking, and sales pipeline stages (New, Contacted, In Progress, Closed).
- **Simple Dashboard System:** Centralized analytics dashboard with metric summary cards (totals, active counts, category breakdown) and recent activity audit logs.

3. Core Laravel Technical Topics to Implement & Document

Your mini project must demonstrate competency across these five core Laravel framework pillars:

1. Migrations, Seeding & Eloquent ORM:

- Define database tables using structured Laravel migration files (no manual SQL importing).
- Provide database seeders to populate initial test records for demo purposes.
- Configure Eloquent Models with mass-assignment protection (\$fillable attributes) and type casting.

2. Routing Architecture:

- Define clean, RESTful endpoints in routes/web.php.
- Use named routes and resource controller routing conventions.

3. **Middleware:**

- Implement at least one custom middleware (e.g., authentication check, session status check, or role guard).
- Ensure unauthorized users are intercepted and redirected with flash warning messages before accessing protected routes.

4. **Controllers & Blade Views:**

- Use a Resource Controller to handle CRUD operations (listing, creating, updating, deleting).
- Build a unified master layout using Blade template inheritance (@extends, @yield, @section) with a shared navigation bar.
- Use @forelse directives for table rendering with empty-state handling, along with pagination.

5. **Server-Side Form Validation:**

- Enforce server-side validation rules on every form (e.g., required, numeric, min/max limits, unique records).
- Maintain 'sticky' forms using the old() helper to retain input values when validation fails.
- Display clear, inline validation error messages next to fields using @error directives.
- Include CSRF protection tokens (@csrf) in all submission forms.

4. **Required Paper Documentation Outline (3 to 5 Pages)**

Your written report must follow this concise structure:

- **Part 1: Project Overview & Scope:** System title, group member names, and a one-paragraph summary of what the system does and who uses it.
- **Part 2: Database Table Structure:** A clear Data Dictionary table displaying column names, data types (string, integer, decimal, boolean), constraints (primary key, unique, nullable), and short functional descriptions.
- **Part 3: Route & Security Summary:** A summary table listing the system's URL endpoints, HTTP methods (GET, POST, PUT, DELETE), controller actions called, and middleware protection status.

- **Part 4: Form Validation Rules Matrix:** A specification table detailing each input field and the corresponding server-side validation rules applied.
- **Part 5: User Interface Screenshots:** Annotated screenshots of the working web application:
 - a. Main record listing table (with data and pagination).
 - b. Create / Add new record form.
 - c. Form validation failure state (highlighting error messages and sticky inputs).
 - d. Success flash notification alert after adding/updating a record.

5. Midterm Grading Rubric (100 Points Total)

Assessment Category	Criteria & Documentation Evidence	Points
Database & Models	Working migrations, automated seeding with sample records, normalized schema, and secured Eloquent models.	20 pts ▾
Routing & Middleware	RESTful resource routes, named routes, and effective custom middleware security.	20 pts ▾
Controller & Form Validation	Complete CRUD operations, server-side validation rules, sticky input handling, and clear error messages.	25 pts ▾
Blade Layout & UI Design	Master layout inheritance, clean presentation,	15 pts ▾

Assessment Category	Criteria & Documentation Evidence	Points
	pagination, CSRF protection, and flash messages.	
Paper Documentation & Demo	Complete 5-part paper report with screenshots, data dictionary, and clear oral demonstration.	20 pts ▾
Total		100 pts ▾